

Physics, 2018 EC, entrance exam, version 2

Grade 12 · Physics · 2018 E.C.

Time allowed: 120 minutes

Questions: 55

Total marks: 55

Name _____ Grade _____ ID No. _____ Date _____

1. A train moving in the east direction accelerates from rest to 36km/h in 20s. What is the average acceleration during that time interval? (1)
 - A. 1.8m/s^2 east
 - B. 0.5m/s^2 east
 - C. 18 m/s^2 east
 - D. 5m/s^2 east
2. The motion of an object at constant speed in straight line is known as_____? (1)
 - A. Uniformly accelerated motion
 - B. Uniform motion
 - C. Accelerated motion
 - D. Decelerated motion
3. What does the slope of velocity–time graph indicates? (1)
 - A. Displacement
 - B. Acceleration
 - C. Speed
 - D. Velocity
4. Work is defined as: (1)
 - A. Force times distance in the direction of the force
 - B. Mass times acceleration
 - C. Energy transferred over time
 - D. Power times time
5. If a machine does 200 J of work in 10 seconds, what is its power output? (1)
 - A. 20 W
 - B. 50 W
 - C. 100 W
 - D. 200 W
6. The work-energy theorem states that: (1)
 - A. Work done on an object equals its change in momentum
 - B. Work done on an object equals its change in kinetic energy
 - C. Work done on an object equals its change in potential energy
 - D. None of the above
7. An astronaut weighs 600 N on Earth. What would be their weight on a planet where gravity is $\frac{1}{4}$ that of Earth's? (1)
 - A. 150 N
 - B. 300 N
 - C. 600 N
 - D. 1200 N
8. When an object rolls down an incline, what type of energy conversion occurs? (1)
 - A. Kinetic to potential energy
 - B. Potential to kinetic energy
 - C. Mechanical to electrical
 - D. Electrical to mechanical

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9. A train is traveling on a straight track at a constant speed. In 80 seconds it covers a distance of 2400 meters. What is the speed of the train? (1)
- A. 30m/s
B. 40m/s
C. 60m/s
D. 100m/s
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10. Pool balls remain at rest until an unbalanced force is applied to them. After they are in motion, they stay in motion until another force opposes their motion relative to an inertial frame with constant velocity. Which laws of motion describes this statement? (1)
- A. Newton's third law
B. Newton's second law
C. Newton's laws of universal gravitation
D. Newton's first law
-
11. A kind of friction that exists between two surfaces in contact when one body tends to slide over the other without moving. (1)
- A. Limiting friction
B. Kinetic friction
C. Static friction
D. Dynamic friction
-
12. Which one of the following objects does not have gravitational potential energy? (1)
- A. A bottle on the surface of a table
B. A ball on the level ground
C. A stone on the roof of a house
D. A fruit on a branch of a tree
-
13. Which one contains the set of vectors? (1)
- A. Velocity, force and acceleration
B. Speed, velocity and time
C. Distance, displacement and speed
D. Mass, time and speed
-
14. Vector resolution is the process of: (1)
- A. Adding multiple vectors together
B. Finding the components of a vector along specified axes
C. Multiplying a vector by a scalar
D. Subtracting vectors
-
15. A force of 10 N acts at an angle of 30° to the horizontal. Find the horizontal component of this force. (1)
- A. 5N
B. 7.07 N
C. 8.7 N
D. 14.14 N
-
16. The instantaneous velocity of an object at any point in time can be determined from the position-time graph by calculating the: (1)
- A. Area under the curve
B. Slope of the tangent at that point
C. Y-intercept
D. X-intercept
-
17. A ball is thrown upwards with an initial velocity of 15 m/s. How high will it go? (Assume $g = 10 \text{ m/s}^2$) (1)
- A. 12 Meters
B. 11.25 meters
C. 225 meters
D. 14.18 meters

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18. A copper wire and a steel wire of the same diameter and length are connected end to end and a force is applied, which stretches their combined length by 1cm. The two wires will have (1)
- A. Different stresses and strains
B. The same strain but different stresses
C. The same stress and strain
D. The same stress but different strains
-
19. The area of cross section of a steel wire ($Y = 2.0 \times 10^{11} \text{ N/m}^2$) is 0.1 cm^2 . What force is required to double its length? (1)
- A. $2 \times 10^2 \text{ N}$
B. $2 \times 10^6 \text{ N}$
C. $2 \times 10^4 \text{ N}$
D. $2 \times 10^9 \text{ N}$
-
20. Which of the following is not correct about Properties of electric field lines? (1)
- A. The field lines never intersect or cross each other
B. The field lines are perpendicular to the surface of the charge
C. The magnitude of charge and the number of field lines are proportional to each other
D. Field lines originate at negative charge and terminate at a positive charge
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21. The potential difference between the terminals of an electric heater is 60V when it draws a current of 4A from the source. What current will the heater draw if the potential difference is increased to 120V? (1)
- A. 4A
B. 3A
C. 6A
D. 8A
-
22. What is the equivalent resistance and the current across 4Ω resistor shown in a Figure below respectively? (1)
- A. 3.2Ω and 7.21A
B. 6.4Ω and 2.7A
C. 7.21Ω and 0.83A
D. 8.3Ω and 22A
-
23. Which of the following modern technologies is a direct application of physics in the field of medicine? (1)
- A. The production of organic fertilizers
B. The classification of plant species
C. The study of ancient languages
D. The development of Magnetic Resonance Imaging (MRI)
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24. What is the primary overall objective of the Ethiopian Physical Society (EPS)? (1)
- A. To sell physics textbooks and laboratory equipment to high schools
B. To replace the Ministry of Education in designing the national curriculum
C. To promote physics education, research, and its applications across Ethiopia
D. To ensure that every student in Ethiopia becomes a professional physicist
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25. Which of the following is the correct order of steps often used in the 'Making of Physics Knowledge'? (1)
- A. Conclusion $\rightarrow$ Hypothesis $\rightarrow$ Observation $\rightarrow$ Experiment
B. Observation $\rightarrow$ Hypothesis $\rightarrow$ Experiment $\rightarrow$ Analysis/Conclusion
C. Experiment $\rightarrow$ Conclusion $\rightarrow$ Theory $\rightarrow$ Observation
D. Theory $\rightarrow$ Law $\rightarrow$ Observation $\rightarrow$ Guess
-
26. Suppose $A = 6i + 4j$ and $B = 2i - cj$ are two vectors in xy-plane. What is the value of c that $A + B = 8i + 2j$? (1)
- A. -2
B. -3
C. 2
D. 3

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27. For what value of b the vectors $A = i + 2j + 3k$ and $B = i + bj - k$ are orthogonal? (1)
- A. 1
B. 2
C. 3
D. 4
-
28. The types of vectors lying on the same plane are (1)
- A. Parallel vectors
B. Collinear vectors
C. Coplanar vectors
D. Orthogonal vectors
-
29. Which of the following is true for an object in uniformly accelerated motion? (1)
- A. The object moves at constant speed
B. The object's velocity changes by the same amount in each time interval
C. The object's acceleration is zero
D. The object moves in a straight line with zero velocity
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30. The coordinate of a particle in meters is given by $x(t) = 18t - 3t^2$, where the time t is in seconds. At what value t will the particle become momentarily at rest? (1)
- A. 6 sec
B. 4 sec
C. 3 sec
D. 0 sec
-
31. A stone dropped from rest at a height h takes time t to reach the ground. If it were dropped from height $9h$, the time to reach the ground is: (1)
- A. $2t$
B. $4t$
C. t
D. $3t$
-
32. A string can withstand a tension of 16 N. What is the greatest speed at which a body of mass 2 kg can be whirled in a horizontal circle using a string of length 2.0m? (1)
- A. 4m/s
B. 5m/s
C. 16m/s
D. 25m/s
-
33. From the following fundamental forces which one is the weakest in strength? (1)
- A. The strong nuclear force
B. The gravitational force
C. The weak nuclear force
D. The electromagnetic forces
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34. A 5 kg block is pulled across a rough surface by a 50 N force at an angle $\phi = 37^\circ$ above the horizontal as shown in figure below. The coefficient of kinetic friction is $\mu_k = 0.4$. Determine the acceleration of the block. (1)
- A. 6.4m/s^2
B. 10m/s^2
C. 8m/s^2
D. 1.6m/s^2
-
35. A pendulum is released from a height of 5 m above its lowest point. At what point in the swing will the pendulum have its maximum kinetic energy? (1)
- A. At the highest point on either side
B. Halfway down
C. At the lowest point
D. At three-quarters of the way down
-
36. A particle moves in a complete circle, starting at Point X and ending at Point X. If the net work done on the particle by a specific force field during this round trip is non-zero, this force is: (1)
- A. Conservative force
B. A non-conservative force
C. A restorative force
D. An electrostatic force

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- 37.** A 10.0g bullet is fired into a stationary block of wood ($m = 5.00\text{kg}$). The bullet sticks into the block, and the speed of the bullet-plus combination immediately after collision is 0.60m/s. What was the original speed of the bullet? (1)
- A. 0.9m/s
B. 0.012m/s
C. 300.6m/s
D. 30 m/s
-
- 38.** When a beam of light falls on an object from the source of light then light get reflected in all direction after striking the objects. The reflected light reach our eyes and our brain interprets finally we are able to see object around us. Which physics branch concern this action? (1)
- A. Mechanics
B. Astrophysics
C. Acoustic
D. Optics
-
- 39.** Civil engineering relies heavily on principles of mechanics to design structures like bridges and buildings. Understanding which concept is crucial for ensuring the stability of a bridge? (1)
- A. Electromagnetism
B. Fluid dynamics
C. Statics and dynamics equilibrium
D. Thermodynamics
-
- 40.** What principle does ultrasound imaging rely on to generate image? (1)
- A. Refraction of sound wave
B. Reflection of sound wave
C. Absorption of sound wave
D. Diffraction of sound wave
-
- 41.** An Ethiopian defense force sends a signal to a target. The signal takes 0.01 seconds to return. What is the distance between the radar and the target in meter? (1)
- A. 3×10^6
B. 3×10^{10}
C. 1.5×10^6
D. 1.5×10^{10}
-
- 42.** Which of the following pair angles have equal range? (1)
- A. 15° and 60°
B. 30° and 80°
C. 20° and 60°
D. 25° and 65°
-
- 43.** A 0.2kg red ball is thrown horizontally at a speed of 4m/s from the height of 3m. A 0.4kg green ball is thrown horizontally from the same height at the same time with a speed of 8m/s. Which ball hits the ground first? (neglecting air resistance) (1)
- A. The red ball hits the ground first
B. The green ball hits the ground first
C. Both strikes the ground at the same time
D. Their time depends on their density
-
- 44.** The launching speed of a certain projectile is two times the speed it has at its maximum height. What is the elevation angle at launching? (1)
- A. 30°
B. 45°
C. 60°
D. 90°
-
- 45.** When a wheel of radius R rotates about a fixed axis, all points on the wheel have (1)
- A. The same linear speed
B. The same angular speed
C. The same linear acceleration
D. Different angular acceleration

46. A motor capable of producing a constant torque 100 Nm and a maximum rotation speed of 200 rad/s is connected to a flywheel with rotational inertia 0.2 kgm^2 . How long will the flywheel take to reach the maximum speed if it starts from rest? (1)

A. 0.4 sec
B. 2 sec
C. 5 sec
D. 500 sec

47. If the semi-major axis of a planet orbit is 9 times the semi-major axis of Earth orbit, what is the approximate orbital period in Earth years? Orbital period of Earth is 1 year. (1)

A. 27 years
B. 54 years
C. 81 years
D. 9 years

48. A satellite is orbiting above the surface of the earth at three times the radius of the earth. What will be the value of the gravitational acceleration 'g' for mass of Earth M and radius of Earth is R? (1)

A. $3GM/R^2$
B. $GM/3R^2$
C. $16GM/R^2$
D. $GM/16R^2$

49. What happens to the gravitational force between two objects if the distance between them is halved? (1)

A. It doubles
B. It remains the same
C. It increases by four times
D. It decreased by four times

50. Why gases easily compress while solids are incompressible? Because (1)

A. Atoms in solids are further apart from each other
B. Atoms in gases are separated by a large distance
C. The force between gas atoms are very strong
D. The force between solid atoms are very weak

51. A substance with specific gravity less than one will be (1)

A. Float in water
B. Sink in water
C. Dissolve in water
D. Break water surface

52. Pressure in liquid depends on (1)

A. Shape of the container
B. Area of the container
C. Density of liquid and depth
D. Volume of the container

53. At a given depth in a static fluid, the pressure acts: (1)

A. Only downward
B. Only upward
C. Equally in all directions
D. Only against the walls of the container

54. Pascal's principle states that pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid. This principle is the basis for: (1)

A. Airplanes
B. Hydraulic lifts
C. Steam engines
D. Thermometers

55. A small piston of a hydraulic lift has an area of 0.002 m^2 . A car weighing 20000N sits on a rack mounted on the large piston. The large piston has an area of 0.8 m^2 . How large force must be applied to the small piston to support the car? (1)

A. 5N
B. 50N
C. $1.6 \times 10^4 \text{ N}$
D. $8 \times 10^6 \text{ N}$

— END OF EXAMINATION —

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